

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the project.	Yes	4-space indentation is maintained consistently in Python files and HTML templates.
2	Proper File Structure	Files and folders are organized logically.	Yes	The project is organized into templates , static , model , and the main app.py file.
3	Meaningful Variable Names	Variable names clearly describe their purpose.	Yes	Variables such as ApplicantIncome, LoanAmount, Credit_History, and Property_Area are self-explanatory.
4	Function / Method Names	Functions have descriptive names.	Yes	Functions such as home(), predict(), and submit() clearly indicate their functionality
5	Code Comments	Comments explain important sections of the code.	Yes	Comments are added for preprocessing, model loading, and prediction logic.
6	Modular Design	Code is divided into reusable functions or modules.	Partial	The preprocessing and prediction logic are modular, but utility functions can be separated into a dedicated module in future versions.
7	No Redundant Code	Duplicate or unnecessary code is avoided.	Yes	Repeated code is minimized by using reusable functions and shared templates.

8	Error Handling	Exceptions are handled appropriately.	Partial	Basic input validation is available. Additional exception handling can be implemented for invalid user inputs and prediction failures.
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Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Functions	Python	Used before every prediction request to preprocess applicant details.	High
2	Random Forest Model (smart_lender_model.pkl)	Scikit-learn / Pickle	Loaded once and reused for every prediction.	High
3	HTML Templates	HTML, Jinja2	Shared layout used across home.html, predict.html, and submit.html.	High
4	CSS Stylesheet	CSS3	Common styling applied across all web pages.	High
5	Flask Routes	Flask	Routes handle navigation and prediction workflow consistently.	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Project files are organized into separate folders for templates, static resources, model, and application logic.
Readability	4	Descriptive variable names, meaningful function names, and comments improve code readability.
Reusability	4	Common preprocessing functions, templates, and the trained model are reused throughout the application.
Documentation / Comments	3	Important sections are documented with comments. A detailed README file can further improve documentation.
Overall Score	4	The project follows a clean and maintainable structure with good readability and reusable components, making it easy to extend in future versions.